

KAPFENBERG, Austria

WMO: 111760

Lat: 47.458N

Lon: 15.336E

Elev: 517

StdP: 95.27

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.9	-14.5	1.1	-10.9	-12.4	1.4	-7.9	5.1	4.0	4.2	2.4	0.8	80	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	30.7	20.0	28.9	19.5	27.1	18.7	21.0	28.4	20.1	27.0	19.3	25.4	1.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.4	23.3	17.8	13.6	22.4	17.2	13.0	21.7	63.1	28.6	59.9	27.1	57.2	25.5	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.2	3.4	2.9	-14.6	33.8	3.3	1.3	-17.0	34.7	-18.9	35.5	-20.8	36.2	-23.2	37.1		
(5)			-14.9	22.7	3.2	0.5	-17.2	23.1	-19.1	23.4	-20.8	23.6	-23.2	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-1.5	1.2	5.3	9.9	14.0	18.2	19.7	19.1	14.6	10.1	4.6	-0.6		
	DBStd	8.21	4.01	3.91	3.85	3.52	3.44	3.52	3.16	3.05	3.39	3.65	3.72	3.74		
	HDD10.0	1344	357	247	148	43	6	1	0	0	3	46	165	328		
	HDD18.3	3359	616	481	403	252	141	44	22	27	118	257	411	586		
	CDD10.0	1199	0	0	3	41	129	246	302	284	142	47	4	0		
	CDD18.3	171	0	0	0	0	6	40	66	52	7	0	0	0		
	CDH23.3	1582	0	0	0	10	87	393	583	456	52	1	0	0		
	CDH26.7	462	0	0	0	0	16	113	185	142	6	0	0	0		
(14) Wind	WSAvg	1.2	1.1	1.2	1.3	1.3	1.4	1.3	1.2	1.2	1.1	1.0	1.0	1.1		
(15) Precipitation	PrecAvg	933	36	38	53	58	99	126	130	127	92	72	58	45		
	PrecMax	1184	75	98	153	123	210	193	277	216	149	163	117	99		
	PrecMin	634	3	2	8	10	42	61	38	55	30	10	0	7		
	PrecStd	120	21	25	29	25	36	35	53	45	29	38	25	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	14.5	19.7	24.9	29.2	32.0	32.9	32.9	27.3	22.2	16.8	10.2		
		MCWB	5.4	7.7	10.7	14.2	18.0	20.2	20.5	20.6	18.5	15.8	11.7	6.9		
	2%	DB	6.7	10.9	17.2	21.7	25.7	29.8	30.6	30.4	24.9	19.3	13.3	7.4		
		MCWB	4.2	6.2	9.9	12.6	16.7	19.6	20.2	20.1	17.5	14.3	10.6	4.5		
	5%	DB	4.9	8.4	14.9	19.7	23.5	27.7	28.7	27.9	22.7	17.1	11.3	5.3		
		MCWB	3.0	4.6	8.7	11.7	15.5	18.7	19.6	19.3	16.3	13.1	9.2	3.7		
	10%	DB	3.4	6.4	12.3	17.4	21.2	25.6	26.8	25.8	20.6	15.1	9.6	4.0		
		MCWB	1.8	3.7	7.5	10.7	14.4	17.7	18.8	18.5	15.3	12.1	7.9	2.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	8.7	11.5	14.8	18.7	21.8	22.1	21.8	19.1	16.5	12.4	7.7		
		MCDB	8.2	13.4	18.1	23.8	26.3	28.9	29.3	29.7	25.3	21.4	15.1	10.0		
	2%	WB	4.6	6.6	10.3	13.2	17.5	20.4	21.1	20.6	18.0	14.9	11.0	5.2		
		MCDB	6.3	10.0	16.3	20.2	24.2	27.5	28.2	28.3	23.4	18.3	13.2	6.4		
	5%	WB	3.3	5.1	9.1	12.2	16.3	19.4	20.3	19.7	17.1	13.6	9.5	3.9		
		MCDB	4.7	8.0	14.1	18.2	21.8	25.8	26.9	26.6	21.7	16.5	11.1	5.0		
	10%	WB	2.0	3.9	7.8	11.2	15.1	18.4	19.4	18.9	16.1	12.5	8.2	2.8		
		MCDB	3.0	6.0	11.8	16.6	19.8	24.1	25.2	24.6	19.5	14.7	9.5	3.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	8.7	11.0	11.6	11.1	11.3	11.5	11.2	10.0	8.8	5.8	5.4		
		MCDBR	7.3	12.2	16.3	16.7	15.8	15.2	15.0	15.2	13.1	11.2	8.0	6.6		
		MCWBR	5.0	7.6	9.0	8.0	7.1	5.8	5.6	5.9	6.1	6.4	5.1	4.7		
		MCWBR	6.7	11.2	15.3	14.6	13.9	13.8	13.7	13.5	11.6	10.7	7.3	6.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.297	0.325	0.368	0.413	0.418	0.424	0.427	0.425	0.394	0.362	0.330	0.297		
		taud	2.394	2.329	2.271	2.207	2.246	2.268	2.261	2.283	2.340	2.402	2.443	2.446		
		Ebn at Noon	786	832	845	838	848	843	833	818	810	780	738	744		
		Edh at Noon	69	91	113	133	133	131	130	122	105	85	67	59		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.20	2.07	3.10	4.12	4.78	5.23	5.22	4.66	3.38	2.32	1.30	1.00		
		RadStd	0.13	0.25	0.34	0.43	0.54	0.40	0.47	0.54	0.43	0.22	0.19	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (109 neighbors)	+0.53	N/A	N/A	+0.87	+0.47	+0.40	-103	-138	+88	+50

Nomenclature: See separate page